

5.3.11

AI25BTECH11017-BALU

Question:

If $\mathbf{A} = \begin{pmatrix} a & b \\ b & a \end{pmatrix}$ and $\mathbf{A}^2 = \begin{pmatrix} \alpha & \beta \\ \beta & \alpha \end{pmatrix}$, then

Solution:

Let us solve the given equation theoretically and then verify the solution computationally
According to the question,

We know that

$$\|\mathbf{A} - \lambda \mathbf{I}\| = 0 \quad (0.1)$$

$$\left\| \begin{pmatrix} a - \lambda & b \\ b & a - \lambda \end{pmatrix} \right\| = 0 \quad (0.2)$$

$$\lambda^2 - 2a\lambda + a^2 - b^2 = 0 \quad (0.3)$$

using Cayley-Hamilton theorem

$$\mathbf{A}^2 - 2a\mathbf{A} + (a^2 - b^2)\mathbf{I} = 0 \quad (0.4)$$

$$\mathbf{A}^2 = 2a\mathbf{A} - (a^2 - b^2)\mathbf{I} \quad (0.5)$$

$$\mathbf{A}^2 = \begin{pmatrix} a^2 + b^2 & 2ab \\ 2ab & a^2 + b^2 \end{pmatrix} \quad (0.6)$$

$$\alpha = a^2 + b^2, \beta = 2ab \quad (0.7)$$